Canonical cutoff-independent coefficients of the $d = 2$ expansion

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Abstract

Astra #8 rank 1. The face finite parts FP^M_γ of the reduced expansion carry a truncation order M; for $\gamma < M \leq M'$ the value is independent of M (`axisFinitePart_indep`), so a canonical finite part at the order $\lceil \max(\gamma, 0)\rceil + 1$ is well defined and equals every admissible one for the analytic faces. On top of this we define canonical coefficient functions of s at an exponent α: $U_\alpha = k_1^{-1}\mathrm{FP}_{k_2\alpha - h_2 - 1}(a_i)$ if $\alpha = \alpha_i$, V_α likewise, $C_\alpha = c_{ij}/(k_1 k_2)$ at collisions, all zero otherwise, and the pole coefficients $A_\alpha = M[\alpha; 0; C_\alpha]$, $B_\alpha = M[\alpha; 0; U_\alpha] + M[\alpha; 0; V_\alpha] - M[\alpha; 1; C_\alpha]$, with $N^{-\alpha}(A_\alpha \log N + B_\alpha)$ equal to the three grouped contributions of the merged sum (`canon_term_eq`). Zero `sorry`/`axiom`.

1 The things formalised

```
theorem axisFinitePart_succ (  b H :  ) (M :  ) (hM :   < M) (hb : 0 < b) (f :   →  )
    (fm :   →  ) (hf : Measurable f)
    (hrem :   v   Ioc (0 :  ) b, |taylorRem f fm M v|   H * v ^ M) :
    axisFinitePart   b f fm (M + 1) = axisFinitePart   b f fm M
theorem axisFinitePart_indep (  b :  ) (M M' :  ) (hM :   < M) (hMM' : M   M') (hb : 0 < b)
    (f :   →  ) (fm :   →  ) (hf : Measurable f) (H :   →  )
    (hrem :   n,   v   Ioc (0 :  ) b, |taylorRem f fm n v|   H n * v ^ n) :
    axisFinitePart   b f fm M' = axisFinitePart   b f fm M
noncomputable def canonicalM (  :  ) :   := max   0  + 1
theorem faceFPCoeff_anaFaceU_canonical ... (k :  ) (i M :  ) (hM :   < M) :
    faceFPCoeff   b k (anaFaceU c b i) (fun m => c (i, m)) M
      = faceFPCoeff   b k (anaFaceU c b i) (fun m => c (i, m)) (canonicalM  )
noncomputable def uIdx (h  k  :  ) (  :  ) :   := (k  :  ) *   - h  - 1
theorem uIdx_uExp (h  k  :  ) (hk  : 0 < k ) (i :  ) : uIdx h  k  (uExp h  k  i) = i
noncomputable def canonU (b :  ) (h  h  k  k  :  ) (a :   →   →   →  )
    (c :   →   →   →  ) (  :  ) :   →   :=
  if uExp h  k  (uIdx h  k   ) =   then
    faceFPCoeff ((k  :  ) *   - h  - 1) b k  (a (uIdx h  k   ))
      (fun m => c (uIdx h  k   ) m) (canonicalM ((k  :  ) *   - h  - 1))
  else fun _ => 0
noncomputable def canonC (h  h  k  k  :  ) (c :   →   →   →  ) (  :  ) :   →   :=
  if uExp h  k  (uIdx h  k   ) =     vExp h  k  (vIdx h  k   ) =   then
    fun s => c (uIdx h  k   ) (vIdx h  k   ) s / ((k  :  ) * k )
  else fun _ => 0
noncomputable def canonA (  :  ) (h  h  k  k  :  ) (c :   →   →   →  ) (  :  ) :   :=
  logMoment     0 (canonC h  h  k  k  c  )
noncomputable def canonB ... (  :  ) :   :=
  logMoment     0 (canonU b h  h  k  k  a c  ) + logMoment     0 (canonV b h  h  k  k  bj c  )
    - logMoment     1 (canonC h  h  k  k  c  )
```

```
theorem canon_term_eq ... (  N :  ) :
    N ^ (- ) * (canonA   h  h  k  k  c   * Real.log N + canonB   b h  h  k  k  a bj c  )
      = N ^ (- ) * logMoment    0 (canonU b h  h  k  k  a c  )
        + N ^ (- ) * logMoment    0 (canonV b h  h  k  k  bj c  )
        + transferTerm    1 (canonC h  h  k  k  c  ) N
theorem canonC_collision ... (i j :  ) (hij : uExp h  k  i = vExp h  k  j) :
    canonC h  h  k  k  c (uExp h  k  i) = fun s => c i j s / ((k  :  ) * k )
```

2 Mathematics

Passing from order M to $M+1$, the Taylor remainder loses the term $f_M v^M$, so the regularised integral changes by $-f_M \int_0^b v^{M-1-\gamma} dv = -f_M\, b^{M-\gamma}/(M-\gamma)$, which is exactly $-f_M\, \mathrm{axisPrim}(\gamma, b, M)$ since $M \neq \gamma$; the counterterm sum gains $+f_M\, \mathrm{axisPrim}(\gamma, b, M)$. Both integrals exist because the order-M weighted remainder is integrable (envelope Hv^M, $\gamma < M$) and $v^{M-1-\gamma}$ is integrable at 0. Induction on $M' \geq M$ then gives independence; the analytic faces supply remainder envelopes at every order (`anaFaceU_rem` with the geometric constant $H(s)\rho^{-i-M}/(1-b/\rho)$). The inverse index $\lfloor k_1\alpha - h_1 - 1 \rfloor_{\mathbb{N}}$ recovers i from α_i exactly, which makes "α is a u-pole" the decidable equation $\alpha_{\lfloor\cdot\rfloor} = \alpha$ and gives integrability-free definitions of $U_\alpha, V_\alpha, C_\alpha$. The pole identity is the explicit form $\mathrm{transferTerm}(\alpha, 1, f) = N^{-\alpha}(\log N\, M[\alpha; 0; f] - M[\alpha; 1; f])$ plus ring.

3 Paper-facing meaning

These are the paper's P_μ data in the N-normalisation ($n = N^2$, $\alpha = 2\mu$, $P_\mu(X) = \frac{1}{2}A_\alpha X + B_\alpha$), defined once, independently of any truncation T, as Astra #8 insisted. The next unit regroups the merged sum by exponent (fibres of the two injective axis maps, with the compatibility inequalities excluding out-of-range collisions), discards retained terms beyond $2T$, and states `twoD_taylor_tree_equal`: for every T, $Z(N) - \sum_{\alpha\in\Lambda_{<2T}} N^{-\alpha}(A_\alpha \log N + B_\alpha) = O(N^{-2T}(1 + \log N))$.

4 Lean notes

- `Nat.ceil` on $\mathbb{R}$ is noncomputable: mark `canonicalM` accordingly.
- `transferTerm_one` already exists in `AxisExpansion` (with the factors in the order $\log N \cdot M$); reuse it.
- After `rw [uIdx_uExp, hij, vIdx_vExp, ← hij]` the collision condition becomes `P    P` with both sides syntactically equal; `if_pos  rfl, rfl` closes the goal and a trailing `simp` reports "no goals".
- `Nat.le_induction` with the `induction M', hMM' using` syntax handles the $M \leq M'$ induction; the cast inequality $M \leq n$ comes from `exact_mod_cast`.
- `intervalIntegral.integral_of_le` plus `integral_rpow (Or.inl hc)` and `Real.zero_rpow` give $\int_0^b v^c = b^{c+1}/(c+1)$ in three rewrites.